



Nebula Full-Stack Case Study Job Immersion Program Platform

Context

Nebula helps students become job-ready through practice-based programs led by experienced professionals.

One of Nebula's key products is the **Job Immersion Program**: a short, practical learning experience where students join a small cohort and work on real-world professional case studies designed by coaches.

The goal of this case study is to build a small but solid full-stack module that allows:

- Coaches to create and manage Job Immersion Programs
- Students to discover available programs and enroll in cohorts
- Coaches to organize sessions across the cohort period
- Students to view their enrolled programs and sessions
- Nebula admins to monitor platform activity

You have **4 calendar days** to complete the case study.

We do not expect you to spend more than **8-12 hours** on it.

We value product judgment, technical clarity, architecture, UX, and ownership more than the number of features shipped.

You are allowed to use AI tools such as ChatGPT, Claude, Cursor, Copilot, or similar tools. However, you must be able to explain how you used them, what you accepted, what you rejected, and how you validated the final implementation.

Product Background Job Immersion Program

A Job Immersion Program is a multi-session, practice-based learning experience designed to expose participants to real professional problem-solving.

Each program is created and led by a coach.

The coach designs practical, domain-specific case studies that simulate real-world work situations.

Examples of programs:

- Investment Banking Analyst Immersion
- Product Manager Case Practice
- Data Analyst Bootcamp
- Quantitative Risk Analyst Immersion
- Strategy Consulting Case Program
- Software Engineering Interview Simulation

Program Structure

A program has:

- A title
- A description
- A target audience
- A professional domain
- A coach
- A number of sessions, between 2 and 4
- A recommended cohort size
- A maximum cohort size, up to 20 participants
- A list of learning outcomes
- Optional take-home tasks called “Explorations”

Each session should last around 45 minutes.

Cohort Structure

A program can have multiple cohorts.

Each cohort has:

- A start date
- An end date
- A maximum number of participants
- A list of scheduled sessions
- An enrollment status

Sessions must be distributed across the cohort period.

For example:

Program: Data Analyst Bootcamp

Cohort: July 1 to July 15

Sessions:

- Session 1: July 1 — Introduction to business problem framing
- Session 2: July 6 — Data analysis case
- Session 3: July 11 — Insight presentation
- Session 4: July 15 — Final feedback and career discussion

Objective

Build a full-stack web application where:

1. A student can create an account or use a mocked student role
2. A student can browse available Job Immersion Programs
3. A student can view program details and available cohorts
4. A student can enroll in a cohort
5. A coach can create a program
6. A coach can create cohorts for that program
7. A coach can schedule 2 to 4 sessions within the cohort period

8. A coach can view enrolled students
9. An admin can view a simple dashboard of platform activity

The result should feel like a real early-stage Nebula product module, not just a technical demo.

Required User Roles

You can implement real authentication or use a mocked role switcher if you want to save time.

Required roles:

Student

A student should be able to:

- Browse available programs
- Filter or search programs by title, domain, or coach
- Open a program detail page
- See available cohorts
- Enroll in one cohort
- View their enrolled programs
- View upcoming sessions for each enrolled cohort

Coach

A coach should be able to:

- Create a new Job Immersion Program
- Edit basic program information
- Create one or several cohorts for a program
- Define the start date and end date of each cohort
- Define the maximum number of participants
- Schedule 2 to 4 sessions within the cohort period
- View the list of students enrolled in each cohort
- Add optional “Explorations” linked to the program or to a session

Admin

An admin should be able to see a dashboard with:

- Number of programs created
- Number of active cohorts
- Number of student enrollments
- Number of coaches
- Latest enrollments
- Upcoming sessions

The admin dashboard can be simple, but it should show that you understand the operating model of the platform.

Functional Requirements

1. Program Creation

A coach can create a program with the following fields:

- Program title
- Description
- Domain
Examples: Finance, Consulting, Data, Product, Software Engineering, Marketing, Entrepreneurship
- Target audience
Examples: Undergraduate students, Master's students, career switchers, junior professionals
- Difficulty level
Examples: Beginner, Intermediate, Advanced
- Number of sessions
Must be between 2 and 4
- Recommended cohort size
Default recommendation: 3 participants
- Maximum cohort size
Must be between 1 and 20
- Learning outcomes
Example: "Learn how to structure a credit risk analysis"
- Program status
Draft / Published / Archived

Only published programs should be visible to students.

2. Cohort Creation

A coach can create a cohort for a published program.

A cohort must have:

- Start date
- End date
- Maximum number of participants
- Sessions
- Enrollment status
Open / Full / Closed

Business rules:

- End date must be after start date
- Cohort capacity must be between 1 and 20
- A cohort must contain the same number of sessions as the program
- Each session date must be between the cohort start date and end date
- A student cannot enroll if the cohort is full

- A student cannot enroll twice in the same cohort

3. Session Scheduling

Each session should have:

- Title
- Description
- Date and time
- Duration, default 45 minutes
- Session order

Example: Session 1, Session 2, Session 3

Sessions should be displayed clearly to both students and coaches.

Bonus: propose an automatic session schedule based on the cohort start date, end date, and number of sessions.

Example: if a cohort runs over 14 days and has 4 sessions, the system proposes evenly distributed dates.

4. Student Enrollment

A student can enroll in an open cohort.

After enrollment, the student should be able to see:

- Program title
- Coach name
- Cohort dates
- Session schedule
- Number of enrolled participants
- Optional explorations or take-home tasks

The student should also see a simple “My Programs” page.

5. Explorations

Explorations are optional take-home tasks assigned by the coach.

An exploration should have:

- Title
- Description
- Linked program or session
- Optional due date

Students enrolled in the cohort should be able to view assigned explorations.

Bonus: allow students to submit a short text response to an exploration.

Bonus: allow coaches to add a short feedback comment on the student response.

6. Admin Dashboard

The admin dashboard should show:

- Total programs
- Published programs
- Active cohorts
- Total enrollments
- Total students
- Total coaches
- Upcoming sessions
- Latest enrollments

This dashboard does not need to be complex.

We are testing whether you can structure useful operational information.

Technical Requirements

You may choose your own stack, but we prefer a modern TypeScript full-stack stack.

Suggested stack:

- Next.js / React
- TypeScript
- Tailwind CSS or equivalent
- PostgreSQL, Supabase, SQLite, Neon, or another simple database
- Prisma, Drizzle, or equivalent ORM
- Server Actions or API routes
- Basic tests with your preferred framework

Your application should include:

- A clear data model
- Input validation
- Basic error handling
- Clean folder structure
- Reusable components
- Clear loading and empty states
- A README explaining how to run the project
- A small seed script or sample data

Deployment is a plus but not mandatory.

AI Usage

You are allowed and encouraged to use AI tools.

However, you must include an `AI_USAGE.md` file explaining:

1. Which AI tools you used
2. What you used them for
3. One example where AI helped you move faster
4. One example where AI gave you something wrong, incomplete, or risky
5. How you reviewed and validated the final code
6. Which parts of the implementation you personally checked most carefully

We are not looking for someone who avoids AI.
We are looking for someone who uses AI like a strong engineer.

Product Notes

Please include a `PRODUCT_NOTES.md` file answering:

1. What did you prioritize and why?
2. What did you intentionally not build?
3. What would you improve with one more week?
4. What would you measure after launching this feature?
5. What would make this feature valuable for Nebula's students and coaches?

Expected Pages

You do not need to follow this exact structure, but a good implementation could include:

Public / Student Side

- `/programs`
Browse available programs
- `/programs/[id]`
Program details and available cohorts
- `/my-programs`
Student enrolled programs and sessions

Coach Side

- `/coach/programs`
Coach program list
- `/coach/programs/new`
Create program
- `/coach/programs/[id]`
Program details and cohorts
- `/coach/cohorts/[id]`
Manage cohort sessions and enrolled students

Admin Side

- `/admin`
Platform dashboard

Minimum Expected Data Model

You are free to adapt it, but your model should probably include:

- User
- Program
- Cohort
- Session
- Enrollment

- Exploration

Optional:

- ExplorationSubmission
- CoachProfile
- StudentProfile

We care more about the quality of your modeling than about matching this exact list.

Deliverables

Please submit to admin@nebulaengage.com :

1. GitHub repository link
2. Setup instructions
3. Vercel Deployed link
4. README.md
5. AI_USAGE.md
6. PRODUCT_NOTES.md

Optional:

- Short Loom or video walkthrough, maximum 5 minutes

Evaluation Criteria

We will evaluate your work based on:

- Product understanding
- Full-stack execution
- Code quality
- Data modeling
- UX clarity
- Business rule implementation
- Security and robustness
- AI usage maturity
- Prioritization
- Communication quality

!We prefer a smaller, clean, coherent implementation over a larger but fragile one.

Good luck.